Skills Summary

Two-Digit Subtraction with Place-Value Models

Use this reference while completing your worksheet or when studying.

Skill 1 — Subtract the tens, then the ones

Model: a two-digit number is ten rods + ones cubes. 58 is 5 rods and 8 cubes.

- Take away the **tens** first: cross out that many rods.
- Then take away the **ones**: cross out that many cubes.
- Read what is left: rods first, then cubes.

Example: $58 - 24$
Step 1. The ones digit of the top number is bigger than the ones digit of the bottom number, so every cube I need is already there and no rod has to be broken.
Step 2. Tens: $50 - 20 = 30$. Ones: $8 - 4 = 4$. Put them together: $30 + 4 =$ **34**.

Try it: $76 - 31$

check: **45**

Skill 2 — Break a ten when there are not enough ones

Rule: if the ones you have are fewer than the ones you must take away, **break one ten rod into ten ones cubes** first. The number does not change: $63 = 60 + 3 = 50 + 13$.

Example: $63 - 27$
Step 1. I must take away 7 ones but only 3 are showing, so I break one ten rod into ten cubes before subtracting.
Step 2. Now $63 = 50 + 13$. Tens: $50 - 20 = 30$. Ones: $13 - 7 = 6$. Together: $30 + 6 =$ **36**.

Try it: $45 - 18$

check: **27**

Skill 3 — Estimate by rounding to the nearest ten

Rule: round each number to the nearest ten, then subtract the rounded numbers. Look at the ones digit: 0–4 rounds down, 5–9 rounds up. An estimate tells you about how big the answer should be.

Example: Estimate $72 - 39$
Step 1. The question asks about how big, not exactly how big, so I round both numbers to friendly tens and subtract those instead.
Step 2. 72 rounds to 70 and 39 rounds to 40, so the estimate is $70 - 40 = \mathbf{30}$.

Try it: Estimate $54 - 27$

check: **20**

(!) Watch out: never turn a column around. In $63 - 29$ the ones column is $3 - 9$, not $9 - 3$. If the top digit is too small, break a ten — do not swap.